

Assignment 3 (Weeks 5 and 6)

Instructions

- This assignment covers material from Weeks 5 and 6.
- Submit a **single PDF file** containing all three parts of the assignment.
- All answers should be your own. Do not copy definitions directly from resources.

Part I: Concepts & Theory (Short Answer)

- Q1.** State the four ACID properties of a transaction. Provide one practical example for each property.
- Q2.** Consider the following SQL statements for a bank transfer:

```
BEGIN;
```

```
UPDATE Accounts  
SET Balance = Balance - 500  
WHERE AccountID = 101;
```

```
UPDATE Accounts  
SET Balance = Balance + 500  
WHERE AccountID = 202;
```

```
COMMIT;
```

What would happen if the database crashes after the first `UPDATE` but before `COMMIT`? In your explanation, differentiate between `COMMIT` and `ROLLBACK` and state under what circumstances each is used.

- Q3.** Suppose two users access the same bank account simultaneously:
1. User A reads the balance.
 2. User B updates the balance and commits.
 3. User A reads the balance again and gets a different value.

What specific concurrency phenomenon has occurred? Which ACID property and transaction concept is meant to control this?

- Q4.** What is a View in SQL and what are two advantages of using one? Furthermore, can data *always* be updated directly through a view? Briefly explain the standard conditions required for a view to be updatable.
- Q5.** What is a Common Table Expression (CTE) and how does it differ from a normal subquery? Briefly explain the difference between a **non-recursive** and a **recursive** CTE, and mention one common use case for a recursive CTE.
- Q6.** What is a Stored Procedure? Mention two primary advantages of using stored procedures over standard individual queries in a database application.

Part II: Practical SQL Application

Reference Table for Q7, Q8, and Q9:

EmpID	Name	Department	Salary
1	Alice	HR	60000
2	Bob	Sales	90000
3	Carol	Sales	85000
4	David	HR	75000
5	Eve	IT	95000

- Q7.** Write the SQL syntax to create a view named **SalesEmployees** that contains only the **EmpID**, **Name** and **Salary** of employees from the Sales department.
- Q8.** Write the explicit SQL statements to safely execute the following as a single transaction:
- Start the transaction.
 - Increase Eve's salary by INR 5,000.
 - Permanently save the changes.
- Q9.** Write a SQL query using a **CTE** to find the names and salaries of employees whose salary is strictly greater than the overall average salary of all employees in the table.

Part III: Long Answer / System Design

- Q10.** An online shopping platform processes thousands of orders every minute. Whenever a customer places an order, the following four operations must occur:
- The ordered quantity is deducted from the inventory.
 - The payment is recorded.
 - A new order entry is created.
 - The customer's order history is updated.

Answer the following:

- A.** Explain why these four operations *must* be executed as a single transaction rather than independent queries.
- B.** Discuss how *each* of the four ACID properties ensures correctness and reliability in this specific scenario.
- C.** Describe a specific, real-world problem that could occur in this scenario if **Atomicity** were not enforced.
- D.** Explain the purpose of **Isolation** in this high-traffic platform and describe one concurrency issue (e.g. Dirty Read or Phantom Read) that could severely impact the inventory or payment tracking without proper isolation levels.